Voice Onset Time in English voiceless stops is affected by following postvocalic liquids and voiceless onsets

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Voice Onset Time in English voiceless stops is affected by following postvocalic liquids and voiceless onsets[a)]

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Voice Onset Time is an important characteristic of stop consonants that plays a large role in perceptual discrimination in many languages, and is widely used in phonetic research. The current paper aims to account for Voice Onset Time variation in English that has defied previously understood phonetic and lexical factors, particularly involving stops that are followed in the word by liquids and voiceless obstruents. 122 Canadian English speakers produced 120 /p/- and /k/-initial words ($n = 17\,533$), and word-initial Voice Onset Time was analyzed. It was found that Voice Onset Time is shorter when the following syllable starts with a voiceless obstruent, and that this effect is mediated by speech rate. Voice Onset Time is also longer before postvocalic liquids, even when they are intervocalic. Voice Onset Time generally decreases through the course of the task, and speakers tend to drift during the course of a word reading task, and this is best accounted for by the residual Voice Onset Time of recently spoken words.

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[BVT] Pages: 2166–2177

I. INTRODUCTION

Voice Onset Time (VOT) is an important characteristic of stop consonants that plays a primary role in discrimination of the voicing contrast in many languages (Lisker and Abramson, 1967). It is widely used as an instrumental measure in phonetic, phonological, and psycholinguistic research, and various factors that influence VOT have been reported. For example, for voiceless stops in English, VOT is sensitive to places of articulation, where dorsal VOT is longer than bilabial/alveolar VOT (Fischer-Jørgensen, 1954), as it is in many other languages (Cho and Ladefoged, 1999). VOT in English is also sensitive to contextual factors such as the height, tenseness, and duration of the following vowel, and the voicing of following coda consonants (Klatt, 1975; Port and Rotunno, 1979). Speaking rate influences VOT as well: faster speaking rates correspond to shorter VOTs (e.g., Miller *et al.*, 1986; Kessinger and Blumstein, 1998; Allen *et al.*, 2003). Various prosodic factors are also known to affect VOT, including stress (Lisker and Abramson, 1967), *F*0 (McCrea and Morris, 2005), pitch accent (Pierrehumbert and Talkin, 1992; Cole *et al.*, 2003), and prosodic domain (Cho and Keating, 2009). In addition to these phonetic/phonological factors, lexical properties (e.g., lexical category, lexical frequency, phonological neighborhood density, and lexical competition) have also been reported to affect VOT (Yao, 2009; Goldinger and Summers, 1989; Baese-Berk and Goldrick, 2009; Kirov and Wilson, 2012; Fox *et al.*, 2015; Buz *et al.*, 2016; Nelson and Wedel, 2017), although the reported lexical effects are smaller than those of most contextual factors. Lastly, women have been reported to produce longer VOTs for voiceless stops (e.g., Swartz, 1992; Ryalls *et al.*, 1997; Whiteside and Irving, 1998; Koenig, 2000), while Smith (1978) found longer VOTs for men for voiced stops.

While much is known about factors affecting VOT in English, it might be more precise to say that much is known about factors affecting VOT in English monosyllabic words. Longer words show patterns that are less well understood. For example, in the course of conducting studies using VOT to investigate speech accommodation (e.g., Nielsen, 2008, 2011; Mielke *et al.*, 2013) it has been clear to us that certain words consistently have long VOT and others consistently have short VOT, to an extent that is not accounted for by known factors, e.g., *coolant* always has a long VOT, even for a word with /k/ and a high tense vowel, and *picture* always has a short VOT, even for a word with /p/ and a short lax vowel.[1] In this paper, we will show that the VOT variation exemplified by these extreme words is structured, that it is not accounted for by previously observed factors, and that it can be accounted for by factors referring to following liquids and following onset consonants.

In addition, we will examine the relative contribution of various factors known to influence VOT. Although many phonetic, lexical, and demographic factors have been shown to influence VOT, their relative importance is not fully understood. Yao (2009) investigated the effect of lexical and contextual factors on the variation of voiceless stop VOT in

[a)]Portions of this work were presented as "Modeling Voice Onset Time in English: Factors and their cross-speaker variability"' at the LabPhon 15 satellite meeting on higher-order structure in speech variability, Cornell University and as "Effects of following onsets on voice onset time in English"' at the 170th Meeting of the Acoustical Society of America, Jacksonville, 2015.

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spontaneous speech produced by two speakers, examining factors such as place of articulation, lexical frequency, phonetic context (category of preceding/following phone), speech rate, and utterance position. The resulting linear regression analysis showed that all these factors indeed affect VOT, and that the effect of lexical frequency is smaller than those of contextual factors. However, the model accounted for less than 20% of the variation in both speakers, suggesting that VOT variation in spontaneous speech is a highly complicated phenomenon that may require more sophisticated modeling.

Recently, Chodroff and Wilson (2017) investigated covariation of VOT among all six English oral stops in both isolated and connected speech, by examining cross-category correlations and ordinal and linear relations among the talker-specific means, and using a mixed-effects model. Their results showed that there are clear linear correlations among stop categories (that are stronger within the same voicing category), and that the wide variability in VOT is highly structured across talkers. That is, a speaker with long VOT for /k/ would produce long VOT for /p/ and /t/. The results of their mixed-effects modeling showed significant main effects for voice, place of articulation, speaking rate, utterance position, and lexical frequency, while the effects of vowel height and vowel tenseness did not reach significance. The effects of speaking rate and utterance position were shown to be greater than the effects of place of articulation and lexical frequency, consistent with Yao (2009).

The strong linear correlations among stop categories reported in Chodroff and Wilson (2017) strengthen our understanding of cross-speaker variability in VOT. However, our understanding of the structure of within-speaker variability is still limited, as the relative strength of various lexical and contextual factors has not been thoroughly investigated. At the same time, Allen *et al*. (2003) showed that the ordering of 17 words within a given talker's VOT distribution was largely consistent across speakers, indicating that within-speaker variability of VOT is highly structured, similar to cross-speaker variability. Given the ubiquity of VOT as an instrumental measure in a wide range of research in linguistics and speech science, it is essential to have a better understanding of how various factors contribute to the phonetic implementation of VOT.

The goal of the current study is to explore structured within-speaker variability of VOT within voiceless categories, by seeking to account for the seemingly inexplicable systematic distribution (i.e., the within-speaker distributions of VOT are stable across individuals) observed in both monosyllabic words (Allen *et al*., 2003) and multisyllabic words (Nielsen, 2011), and by examining the relative contribution of various phonetic/phonological and lexical factors. Further, while large speaker variability in VOT has been reported, the distribution of speaker variability for each factor is not well understood. In the course of addressing these issues, the current study reveals two types of previously unreported significant factors involving following liquids and obstruents, and will present an analysis of English voiceless VOT using linear mixed effects regression.

II. METHODS

A. Participants

Participants were 122 Canadian English speakers in Ottawa, Ontario (38 male; age: mean = 20.7, standard deviation = 3.43). All participants reported normal hearing and normal or corrected-to-normal vision and no documented history of learning difficulties.

B. Materials

The stimuli consisted of 150 words: 120 test words with initial voiceless stops and 30 filler words with initial sonorants. Among the test words, 100 were words beginning with /p/ and 20 were words beginning with /k/. All test words were selected from CELEX2 (Baayen *et al*., 1995) with varying lexical frequency, neighborhood density, following vowel, and length (1–3 syllables for /p/ initial words; 2 syllables for /k/ initial words). All words had high familiarity (>6.0 on the 7-point Hoosier Mental Lexicon scale, Nusbaum *et al*., 1984). While all test words had initial stress, the stress pattern of fillers (i.e., words with initial sonorants) was varied. No words with initial onset clusters or initial voiced stops were included. The word list was identical to the one used in Nielsen (2011), which investigated the pattern of VOT accommodation. We chose to analyze the pattern of VOT variation in these words because they show consistent variation between words that is not accounted for by factors reported in previous studies of English VOT.

C. Procedures

The task was a word-naming paradigm, in which the words in the list were visually presented on a monitor (one at a time, every 2 s), and the participants were seated inside a sound-attenuated booth (Génie Audio, Montreal, Canada) and instructed to read the words aloud into a Shure (Niles, IL) Beta 53 head-mounted omnidirectional microphone which was connected to a Sound Devices (Reedsburg, WI) USBPre microphone preamplifier and A/D converter. Audio was recorded at 44.1 kHz with 16 bits per sample. The words were presented in random order for each participant.[2]

D. Data analysis

VOTs of the 17 533 initial stops were measured semi-automatically and all other segment durations were measured using forced alignment and hand correction. The first pass used the Penn Phonetics Lab Forced Aligner (P2FA; Yuan and Liberman, 2008) (10 ms time resolution), followed by hand correction of aspiration intervals (VOT). New phone models were trained on these recordings, with 5 ms time resolution and separate models for plosive closure/release, affricate stop/fricative intervals, and diphthong nucleus/off-glide. Duration outliers were then hand-corrected again. We quantified speech rate as follows. We included by-subject mean word duration in order to account for general speaking rate differences between subjects. To account for fluctuations in speaking rate within the session, we performed a linear regression with word duration as the dependent variable, and by-subject mean word duration and by-word mean word duration and their interaction as factors. The residuals of this model represent deviations from the word duration that would be expected on the basis of by-subject and by-word means, i.e., the residual is a measure of local speech rate. We included the by-subject means and the residuals as factors in the VOT regression model. We also included an interaction between word duration residual and sex.

III. RESULTS

Figure 1 shows the mean VOT and all segment durations for each word. The gray and white bars on the left side of the figure show the mean VOT for each word (/k/ words are in gray and /p/ words are in white). We expect /k/'s to have longer VOTs than /p/'s, all else being equal. The right side of the figure shows the mean durations of the rest of the segments in the word, shaded by manner of articulation. Aspirated stops have been segmented into closure and aspiration intervals, diphthongs have been segmented into nucleus and off-glide intervals, and affricates have been segmented into stop and fricative intervals. Another familiar pattern that is apparent here is that the words with longer VOTs tend to have longer vowels immediately following the /p/ or /k/, and the words with the shortest VOTs tend to have lax vowels and voiceless coda consonants (which are associated with short vowel duration). The by-word mean VOT values are highly correlated with the by-word mean VOT values from the California speakers in Nielsen's (2011) experiment 1 (adjusted $R^2 = 0.8443$). The extreme words from Nielsen's data are the extremes in our data as well: *coolant*, mentioned in Sec. I, has the longest mean VOT and is in the top 10% of words for 62% of our speakers (based on a single repetition per speaker). The /p/ in *picture* has the lowest mean VOT and is in the bottom 10% of words for 70% of our speakers. We will examine the extent to which this variation is accounted for by previously observed factors.

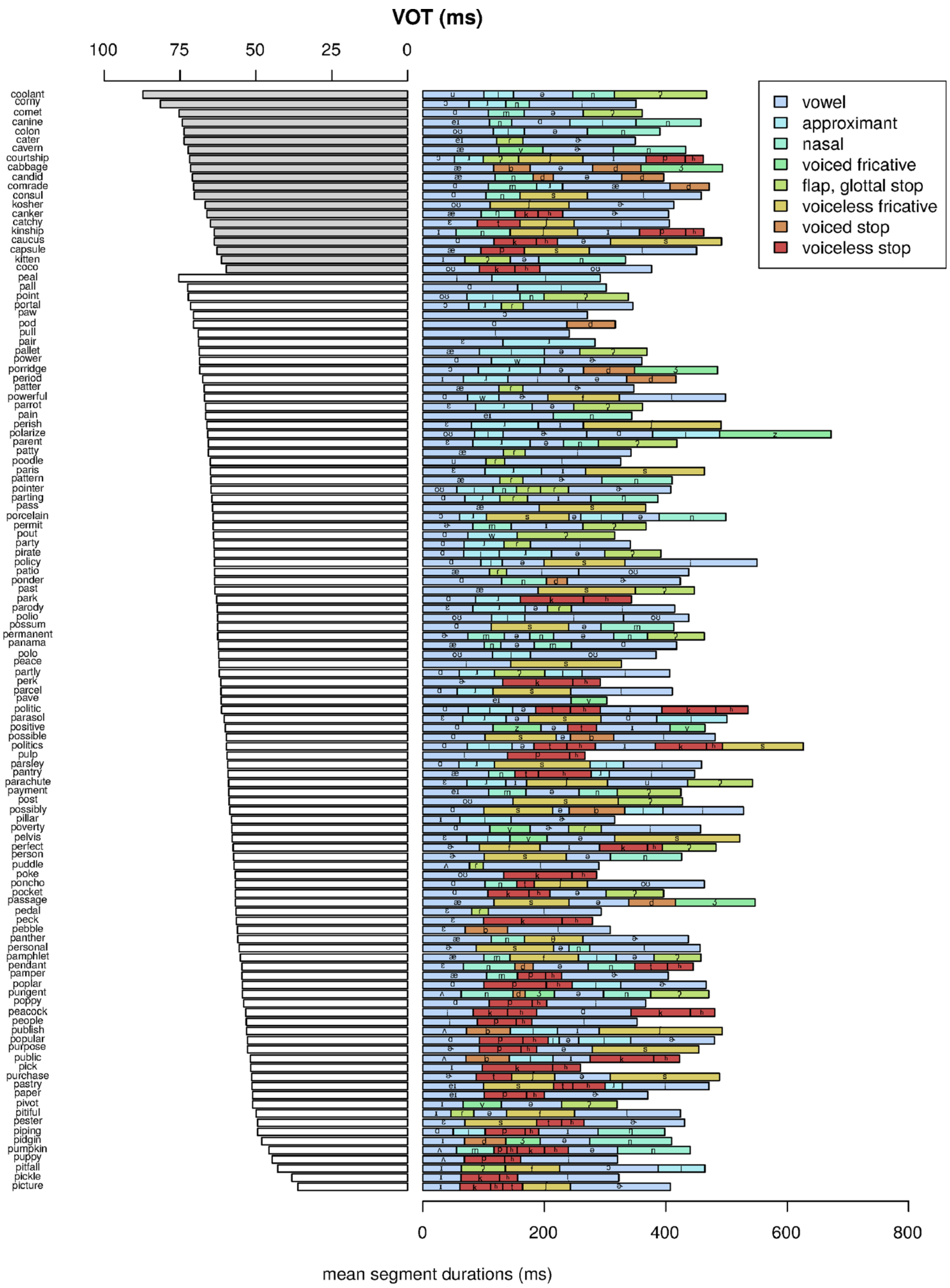

FIG. 1. (Color online) Mean VOT and segment duration by word. Shading reflects manner of articulation. Note that VOT and segment duration are displayed on different scales.

A. Previously observed phonetic, lexical, and demographic factors

We performed a linear mixed effects regression including the following well-established fixed effects: (1) word duration (both residual and by subject), (2) place of articulation, (3) vowel duration (mean by word),[3] (4) vowel tenseness, (5) vowel height and its interaction with tenseness, (6) following voiceless coda consonant, (7) lexical frequency (log CELEX2 frequency, Baayen *et al.*, 1995), (8) neighborhood density (densityB; Sommers, 2002, based on the Hoosier Mental Lexicon; Nusbaum *et al.*, 1984), and (9) speaker sex.[4] Random intercepts were included for speaker and word, and random slopes were included for word duration residual and subject (because the effect of rate has previously been shown to vary across subjects, e.g., Allen *et al.*, 2003). VOT is expressed in milliseconds, and the mean VOT is subtracted from all VOT values (as in Chodroff and Wilson, 2017). All numerical factors were scaled. This model (model 1) is summarized in Table I.

Speech rate has a large effect on VOT, both globally (word duration by subject) and locally (word duration residual). Longer words have a longer VOT as well, and the effect of word and subject duration factors is not additive. Being velar increases VOT, as expected. There is a significant interaction between the tense vowel and high vowel factors, consistent with VOT being longer before high tense vowels such as /i/ and /u/, and being shorter before other high vowels (/ɪ/ and /ʊ/). Words in denser lexical neighborhoods have longer VOTs, but we observe no effect of lexical frequency.[5] By-word vowel duration (the duration of the following vowel, including off-glides of diphthongs /aɪ aʊ ɔɪ/, averaged by lexical item) is only weakly associated with a longer VOT. A following voiceless coda decreases VOT, even though its effect on the preceding vowel is addressed by the vowel duration factor. Males have a longer VOT than females, and they show a larger effect of speech rate (both contrary to previous literature, e.g., Swartz, 1992; Ryalls *et al.*, 1997; Whiteside and Irving, 1998).

Figure 2 shows the same segment durations as Fig. 1, but instead of mean VOTs for each word, it shows the by-word random effects for model 1. Words with positive random effects have VOTs that are longer than expected based on the factors in model 1, and words with negative random effects have VOTs that are shorter than expected. There are two basic observations to make on the basis of this figure. The first observation involves words such as *coolant*, *portal*, *corny*, and others that are at the top of the figure because their mean VOTs exceed the model estimates the most. What these words (and the rest of the top ten words) have in common is that their initial stops are followed by vowel + liquid sequences (which may or may not be in the same syllable). The second observation involves words such as *puppy*, *coco*, *pickle*, *paper*, and others that are at the bottom of the figure because their mean VOTs are substantially shorter than the model estimates. What these words have in common is that the second syllable starts with a voiceless stop or affricate. The presence of a following voiceless stop is common to the 11 words with the most unexpectedly short mean VOTs. There are also words with voiceless fricatives onsets (such as *pitfall* and *panther*) with surprisingly short VOTs, and the word *pave*, which is difficult to interpret because it is the only monosyllabic word in the dataset with a voiced fricative coda. Clearly, following consonants play a role in VOT that is not being properly addressed by model 1.

TABLE I. Summary of model 1 (factors from the literature).

	Estimate	Std. Error	*t* value
(Intercept)	−3.52795	1.45561	−2.424
word duration (residual)	3.30673	0.26023	12.707
word duration (by subject)	9.16622	0.92304	9.930
velar	11.02816	1.43432	7.689
tense vowel × high vowel	12.55602	2.84478	4.414
neighborhood density	2.03182	0.67061	3.030
word duration (residual) × sex	1.21423	0.47436	2.560
sex (male)	5.13940	2.04313	2.515
vowel duration (by word)	1.23737	0.66919	1.849
tense vowel	2.16301	1.21996	1.773
lexical frequency	−0.05461	0.52480	−0.104
voiceless coda	−5.99804	1.65386	−3.627
high vowel	−9.44252	1.98423	−4.759

B. Following voiceless onsets and liquids

A general summary of the observations so far is that word-initial voiceless stop VOTs are sensitive to the manner of articulation of segments beyond the first syllable. We will explore this by trying models with factors for following voiceless consonants and liquids. When trying multiple versions of the same thing, we will select the model with the greatest reduction of the Bayesian information criterion (BIC) from the previous best model. When it is useful to know if one model is significantly better than another, we will use analysis of variance (ANOVA). Since this exercise involves 40 comparisons of potential interest, we will use a significance criterion of 0.05/40 or 0.00125 for these model comparisons.

The VOTs are shorter when there is a following voiceless onset, and they are longer when there is a following liquid. The effect of following voiceless plosive onsets is interesting, because to our knowledge they do not affect the duration of the intervening vowel. This seems to be a direct effect of a consonant on the aspiration of a preceding stop. Incorporating a factor for following voiceless obstruent onset improves model 1, which is the original model shown in Table I and Fig. 2. Model 2a includes the voiceless obstruent onset factor, and model 2b additionally includes a factor for voiceless plosive onsets. Both are significant improvements ($p < 0.0001$), based on pairwise ANOVA comparisons of models, i.e., a voiceless obstruent in the following syllable onset decreases the VOT of the word-initial stop, and if that following obstruent is a stop or affricate, it decreases VOT even more.

How to deal with the vowel + liquid sequences is less obvious, and the options include adding a factor for a following liquid that is independent of the duration measure, or including the duration of a postvocalic liquid in the vowel

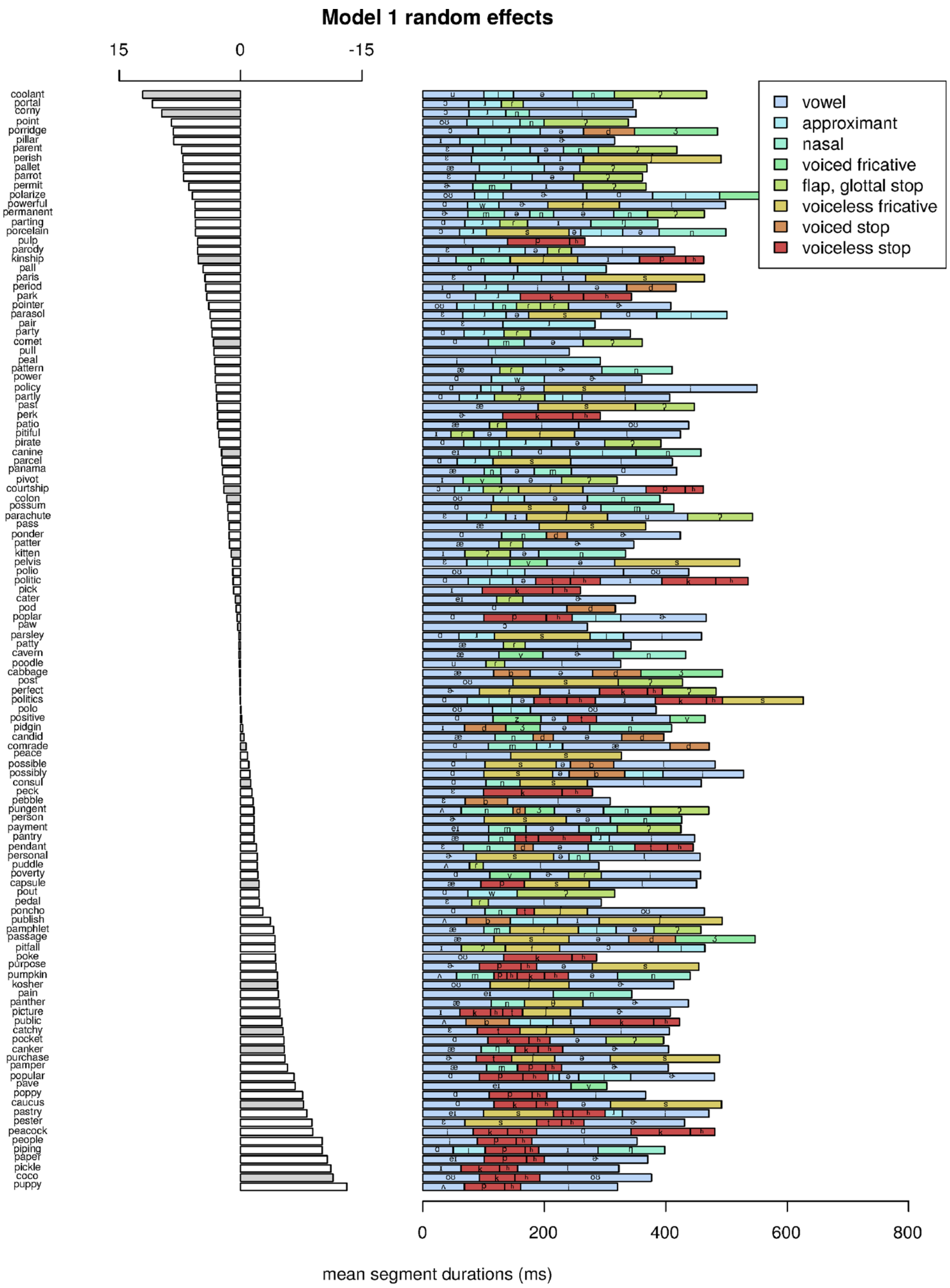

FIG. 2. (Color online) Random effects and segment durations: model 1, which includes factors that are well supported from the previous English VOT literature. Words with positive (left-pointing) random effects have VOTs that are longer than expected based on the factors in model 1, and words with negative random effects have VOTs that are shorter than expected.

duration measure.[6] We tried models with four different factors to represent the presence of a liquid. Model 3a includes a factor for a liquid in the rhyme of the first syllable (e.g., including *portal* but not *period*). Model 3b includes a factor for any syllabic liquid or liquid immediately following the main vowel, including postvocalic onsets (e.g., including *portal* and *period* but not *comrade*). Model 3c includes a factor for any liquid in the rhyme or the following onset, even if not postvocalic (e.g., including *portal*, *period*, and *comrade*). Model 3d includes a factor for a liquid anywhere in the word (e.g., also including *cavern*). We also tried models that handle liquids by incorporating other sonorants in the vowel duration measure instead of adding a new factor for them. Model 3e replaces the original vowel duration measure with one that includes liquids that are in the rhyme (analogous to model 3a). Model 3f instead uses a duration measure that includes all liquids following the vowel (analogous to model 3b). We tried a few other reasonable ways to include liquids in the duration measure. Model 3g's duration measure includes all sonorants in the rhyme, and model 3h's duration measure includes the entire rhyme of the first syllable. Model 3i's duration measure uses the entire duration of contiguous vowels, liquids, and glides following the initial stop, regardless of what syllables those segments appear in. All the models except for model 3d (liquid anywhere in the word) and model 3h (duration includes the entire rhyme) have a lower BIC than model 2b, which these models are building on. Model 3b (which includes a factor for all postvocalic liquids, including onsets) achieves the greatest BIC reduction relative to model 2b. An ANOVA shows it to be significantly better than model 3f (the next best of these models, which includes the same liquids in the vowel duration measure instead of as a separate binary factor) [$p = 0.0001$]. In short, the greatest BIC reduction is achieved by adding a factor representing the presence of any syllabic or postvocalic (including onset) liquid to account for the VOT lengthening effect of following liquids.

C. Interaction with speaker speech rate

The nonadjacent contextual factors (voiceless onset, voiceless plosive onset, and voiceless coda) can conceivably be affected by variation in how close together in time these sounds occur. In model 4a, we included interactions between each of these three phonetic factors and the word duration (by subject) factor, which is a measure of how slowly each subject pronounced the words. Each of these three phonetic factors had negative coefficients in the previous best model 3b (voiceless onset: Estimate = −3.5307, Std. Err. = 0.7854, $t = -4.495$; voiceless plosive onset: Estimate = −4.9118, Std. Err. = 0.9401, $t = -5.225$; voiceless coda: Estimate = −4.7954, Std. Err. = 0.8969, $t = -5.347$). Adding the interaction for voiceless onset × duration (by subject) in model 4a yields a large reduction in BIC. Further adding the interaction for voiceless plosive onset × duration (by subject) in model 4b makes the model worse, and so does adding the interaction for voiceless coda × duration (by subject) in model 4c. As can be seen in the final model below in Table II, the interaction effects have a negative estimate, meaning that the VOT-reducing effects of following voiceless onsets are stronger for speakers who are talking more slowly in general. This suggests that there may be a floor effect in faster speech.

D. VOT variation through the task

Finally we consider how VOT varies during the word-naming task. We tested for two effects: a general trend through the course of the task and fluctuation during the task. Adding a factor for list position (in model 5a) shows that VOT generally declines modestly through the course of the task (Estimate = −0.2834, Std. Err. = 0.1065, $t = -2.662$). Adding the list position factor does not yield a reduction in BIC, but we will keep the factor because it makes the next step easier to interpret.

To test for drifting up and down during the task, we included the residual for the preceding word as a factor, i.e., how unexpectedly long or short the previous target word's VOT was, after taking into account all the factors in model 5a.[7] Adding the residual for the preceding word as a factor improved the model substantially (model 5b, ΔBIC = −143). To test for longer-term effects, we included the residual of the word two target words prior (model 5c) and three target words prior (model 5d), both of which were improvements. Adding a factor for four target words prior was not an improvement (model 5e). Using previous VOTs (z-scored by subject) instead of previous residuals (model 5f) is also a significant improvement on model 5a ($p < 0.0001$), but it is worse than using residuals ($p < 0.0001$). In other words, taking into account the VOT vs expectation for recent words is a better predictor of future VOT than the actual VOT of recent words.

TABLE II. Summary of model 5d (model 1 plus factors accounting for following liquids and voiceless onsets, and short-term VOT drift).

	Estimate	Std. Error	t value
(Intercept)	−4.7786	1.2686	−3.767
velar	11.7948	0.7767	15.185
word duration (residual)	3.2761	0.2547	12.860
Model 5a residual of word $i-1$	1.2141	0.1053	11.533
word duration (by subject)	9.7179	0.9210	10.551
syllabic or postvocalic liquid	5.0736	0.6310	8.041
tense vowel × high vowel	9.5816	1.5930	6.015
tense vowel	3.5354	0.6703	5.274
Model 5a residual of word $i-2$	0.5351	0.1058	5.059
Model 5a residual of word $i-3$	0.5221	0.1053	4.959
vowel duration (by word)	1.6042	0.3791	4.231
neighborhood density	1.1362	0.3647	3.115
word duration (residual) × sex	1.1760	0.4641	2.534
sex (male)	5.1333	2.0352	2.522
lexical frequency	−0.1713	0.2844	−0.602
word list position	−0.2789	0.1057	−2.639
voiceless onset	−3.4418	0.7831	−4.395
voiceless plosive onset	−4.9468	0.9373	−5.278
voiceless coda	−4.8004	0.8942	−5.368
high vowel	−6.2905	1.1261	−5.586
voiceless onset × word duration (by subject)	−1.6822	0.2218	−7.584

E. Summary of best model

Model 5d is our best model, and it is summarized in Table II. We have observed that VOT is longer when there is a following syllabic or postvocalic liquid, even if that liquid is also prevocalic (=onset). In addition to being shorter before voiceless codas, VOT is also shorter before following voiceless onsets, especially if they are stops or affricates, and particularly for speakers with longer word durations, i.e., this effect is diminished for speakers who say the words quickly.

We have also observed that VOT declines through the course of the task, and that short-term fluctuations are more important: the residuals for the three previous target words are all positively correlated with the VOT of the next word, and the residuals are better predictors than the normalized VOT. This means that VOT is longer after words produced with longer than expected VOT, and this is a better predictor than simply knowing whether preceding VOTs are long or short. Knowing whether preceding VOTs are long or short is not as important as knowing whether they are longer or shorter than expected given other factors that account for VOT.

Figure 3 shows the by-word random effects and segment durations for model 5d. The sizes of the random effects are smaller than for model 1 (Fig. 2), because the effects of following liquids and voiceless onsets have been addressed. The words with the largest positive random effects (the words whose VOT is underestimated by the model) include words such as *point*, *kinship*, *pantry*, and *portal*, which have nasals or sequences of sonorant consonants following the stop, whose combined effect is apparently not fully accounted for by model 5d. The words with the most negative random effects include *pave*, which is the only word in the set with a voiced fricative coda. The VOT of *pave* is shorter than expected based on the length of its vowel, but it is difficult to

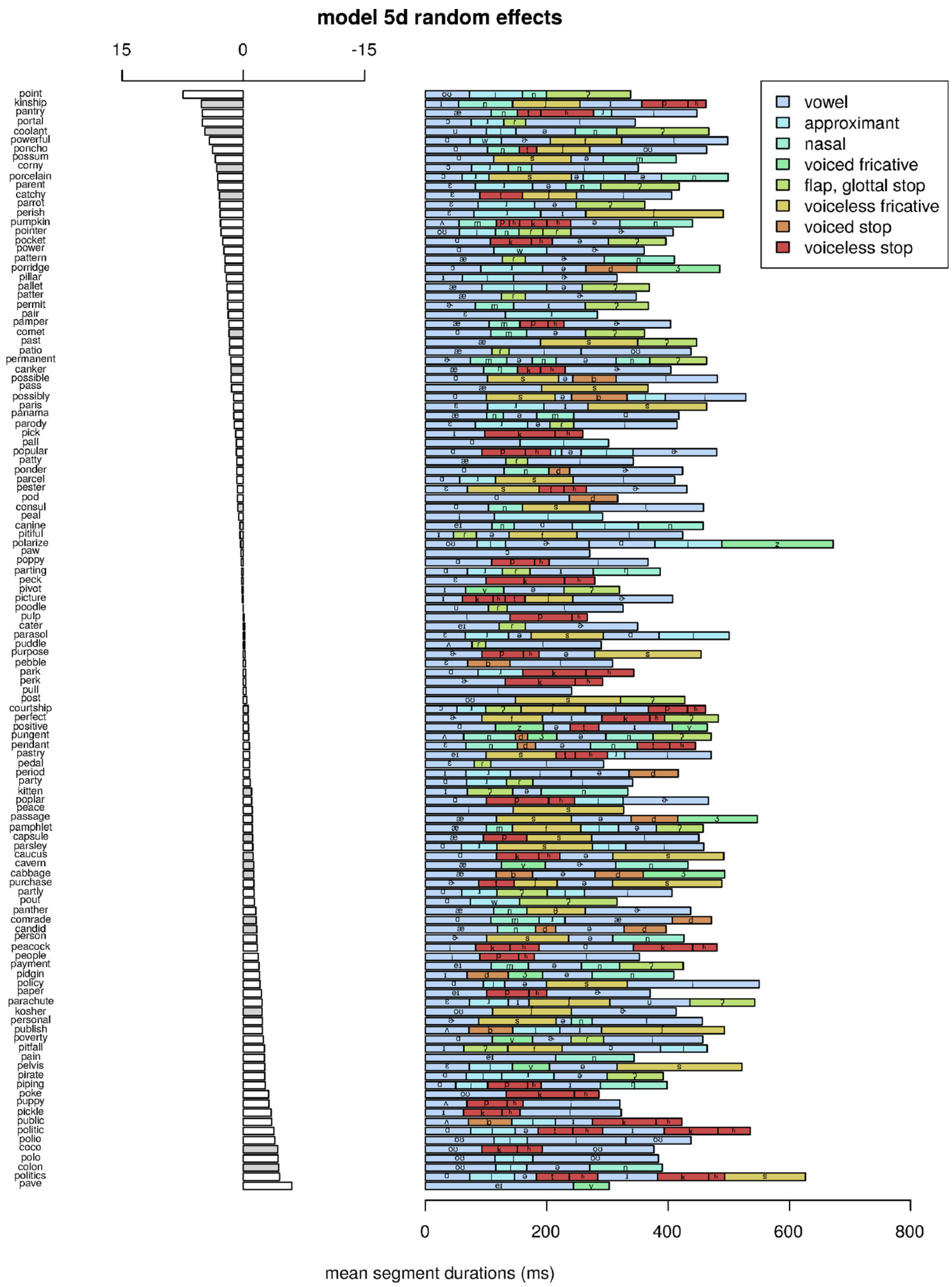

FIG. 3. (Color online) Random effects and segment durations: model 5d, which is model 1 plus factors accounting for following liquids and voiceless onsets, and short-term VOT drift. Words with positive random effects have VOTs that are longer than expected based on model 5d, and words with negative random effects have VOTs that are shorter than expected.

improve our understanding of it without data from more words that are similar to it.

The comparison of model 1 (the model motivated by the previous literature) and model 5d (our best model, accounting for following liquids and onsets, and variation during the task) shows ΔBIC of −341. Table III summarizes all the steps leading from model 1 to model 5d, including factors that did not make it into the final model. The models that directly lead to the best model are bolded in the table.

F. Considering alternate versions of phonetic factors

To this point we have excluded glottalized /t/ and flapped /t/ from the voiceless coda and voiceless plosive onset categories, respectively. Including coda /t/ as a voiceless coda makes model 5d worse (ΔBIC = +11), and including foot-medial /t/ as a voiceless onset and plosive make model 5d worse as well (ΔBIC = +63). This suggests that it is the presence of a phonetically voiceless stop, not the presence of the phoneme /t/, that reduces VOT of a preceding /p/ or /k/.

It is reasonable to expect that longer durations of /t/'s that can be produced as a flap or a glottal stop (which could indicate the production of a full stop) could predict shorter initial stop VOTs. We performed a model with the same factors as model 5d on the subset of words in which the second syllable begins with an underlying /t/ that can be realized as a flap (*cater*, *parting*, *party*, *patio*, *patter*, *pattern*, *patty*, *pitiful*, *pointer*, and *portal*), and compared this with a model which includes the duration of the /t/ as a factor. Adding the duration factor yields a slightly worse model (ΔBIC = +2.5). Longer following onset /t/'s are associated with shorter word-initial onset VOT (Estimate = −1.5571, Std. Err. = 0.7166, $t = -2.173$). A similar model comparison limited to words with glottalizable first-syllable coda /t/'s (*courtship*, *partly*, *pitfall*, *point*, *pout*) found that adding the /t/'s duration as a

TABLE III. Summary of regression models. The models on which the best model is built are in bold. The ΔBIC values are the result of comparing each model with the model referenced in its description column. Model 5d is the best model.

Model	Description	k	BIC	ΔBIC	logLik
1	See Table I	18	116 164	—	−57 996
2a	Model 1 + voiceless onset factor	19	116 107	−57	−57 962
2b	2a + voiceless plosive onset factor	20	116 086	−21	−57 947
3a	2b + factor for liquid in rhyme	21	116 072	−14	−57 936
3b	2b + factor for liquid in rhyme or postvocalic onset	21	116 044	−42	−57 922
3c	2b + factor for liquid in rhyme or next onset (any)	21	116 067	−19	−57 933
3d	2b + factor for liquid anywhere in word	21	116 089	3	−57 944
3e	2b but duration includes liquid in rhyme	20	116 074	−12	−57 941
3f	2b but duration includes postvocalic liquid	20	116 050	−36	−57 929
3g	2b but duration includes sonorants in rhyme	20	116 081	−5	−57 945
3h	2b but duration includes entire rhyme	20	116 087	1	−57 948
3i	2b but duration includes sonorant span	20	116 064	−22	−57 936
4a	3b + voiceless onset × duration (by subj)	22	115 999	−45	−57 894
4b	4a + voiceless plosive onset × duration (by subj)	23	116 009	10	−57 894
4c	4a + voiceless coda × duration (by subj)	23	116 006	7	−57 893
5a	4a + stimulus position	23	116 002	3	−57 891
5b	5a + residual of word i - 1 (previous word)	24	115 859	−143	−57 815
5c	5b + residual of word i - 2	25	115 838	−21	−57 798
5d	5c + residual of word i - 3	26	115 823	−15	−57 787
5e	5d + residual of word i - 4	27	115 831	8	−57 786
5f	5d but using normalized 3 previous VOTs	26	115 858	35	−57 804

factor also yields a slightly worse model (ΔBIC = +4.4), although the effect of duration is in the same direction as with flappable onsets (Estimate = −1.25253, Std. Err. = 0.87141, t = −1.437).

To this point we have used a version of [+tense] which includes low vowels and diphthongs with a low nucleus (following the previous VOT literature). Excluding these vowels from the [+tense] class (and limiting [tense] to /i u eɪ oʊ/) makes model 5d worse (ΔBIC = +25), and so does treating [tense] as a ternary feature in which low vowels occupy a middle category (ΔBIC = +25).

The vowel duration factor in model 5d is an average across all the tokens of a word. It is reasonable to expect that VOT could be positively correlated with vowel duration on a token-by-token basis instead. However, replacing the vowel duration factor with the actual duration of each token reverses the direction of the effect. This is consistent with the idea that longer VOTs come at the expense of vowel duration (i.e., longer VOT means that more of the interval between the stop release and the end of the vowel is voiceless), so while words with longer vowels have initial stops with longer VOTs in general, tokens with shorter vowels are more likely to have initial stops with longer VOTs (Estimate = −41.9197, Std. Err. = 4.5342, t = −9.245 for vowel duration by token, vs Estimate = 1.6042, Std. Err. = 0.3791, t = 4.231 for duration by word in model 5d).

G. Considering monosyllabic words separately

Much of the variation in vowel duration involves the monosyllabic words, where the vowels following initial stops are affected by final lengthening. Adding a factor for monosyllabicity to model 5d does not improve it (ΔBIC = +9), presumably because the effect of monosyllabicity is mediated by vowel duration, which is already in the model. To see if the difference between monosyllabic and bi-/trisyllabic words is the main factor driving the duration effect, we did a regression with the same factors as model 5d but excluded the monosyllabic words. Vowel duration (by word) is significant within the bisyllabic/trisyllabic group, with a slightly larger estimate (Estimate = 2.2596, Std. Err. = 0.6543, t = 3.453). Within the monosyllabic words, the vowel duration (by word) factor is not significant (Estimate = 0.52805, Std. Err. = 0.59588, t = 0.886). This shows that the vowel duration effect is not simply driven by the presence of monosyllabic words, and that in monosyllabic words, the effect of differences in vowel duration is eliminated by final lengthening.

H. Variation between speakers

Adding a random slope for most phonetic and lexical factors improves model 5d, but the model does not converge with all of them in it at the same time.[8] To explore variation between speakers further, we performed a separate linear regression for each speaker and compared the factor estimates. Figure 4 shows the distributions of those estimates. The effects of some factors, such as place of articulation (velar) and vowel height, show greater cross-speaker variability, while the effects of other factors are relatively consistent across speakers (e.g., vowel tenseness). The estimates for lexical frequency (centered around zero) are also less variable compared to those of neighborhood density.

IV. DISCUSSION

This investigation of English VOT showed that VOT is increased by the presence of postvocalic liquids, even those that are typically analyzed as onsets. This accounts for the long VOT in words like *coolant* and *portal*. The estimated

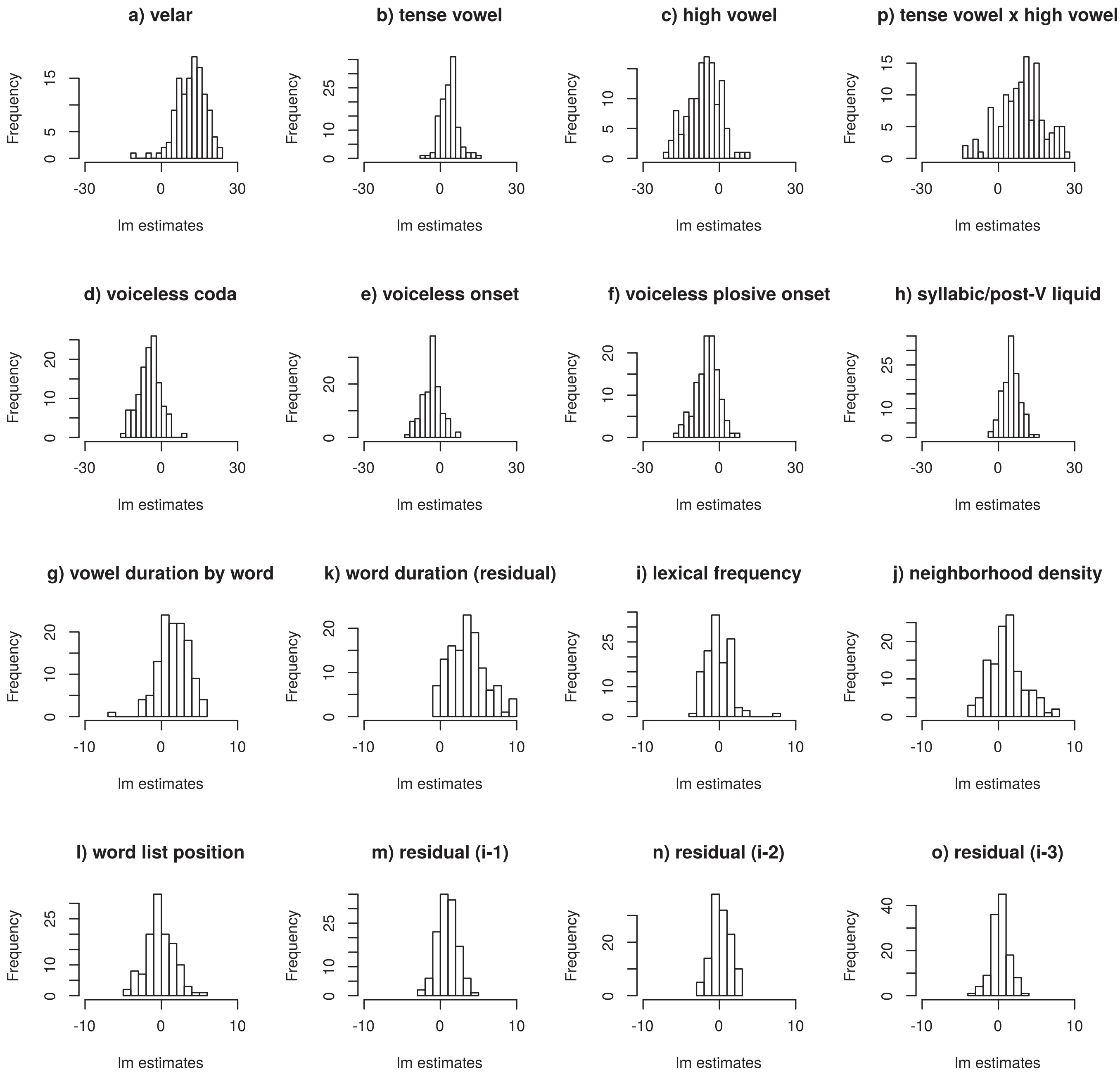

FIG. 4. Estimates from separate linear regressions performed for each speaker. Categorical factors all have x axis ranges of $[-30, 30]$, and (scaled) continuous factors all have x axis ranges of $[-10, 10]$.

effect size for the liquid factor is greater than the effect of following vowel tenseness, but smaller than the effect of being a velar consonant. While the presence of a liquid accounts for observed long VOTs, we have seen that VOT is shorter when the next syllable starts with a phonetically voiceless obstruent, and even shorter when that voiceless obstruent is a stop or affricate. The estimated combined effect of these three factors (as in the word *picture*, which has a voiceless coda followed by a voiceless obstruent which is a plosive) is comparable in magnitude (but opposite in direction) of the effect of being a velar. These effects of following voiceless obstruents and liquids help explain how the /k/ in *coolant* could have a mean VOT that is 2.4 times longer than the mean VOT of the /p/ in *picture* (87 ms vs 36 ms), despite the fact that both are considered to belong to the phonetic category of voiceless stops. They also help account for why the /p/ in *portal*, which is followed in the word by a postvocalic liquid and a flap, has a longer VOT than the /k/'s in words like *capsule* and *caucus*, which are followed in the words by phonetically voiceless obstruents.

The effect of following voiceless consonants is limited to consonants that typically are phonetically voiceless: including underlying /t/ in flapping and glottalizing contexts with voiceless plosives or codas both made the models worse. However, within words with typically flapped following onset /t/s, a longer duration of the /t/ interval was associated with a shorter VOT in the initial stop. This suggests that when usually flapped /t/s are produced as voiceless stops, they exhibit the same inhibiting effect as other voiceless obstruents.

The results also support earlier findings that VOT is generally longer in /k/, directly related to following vowel duration, inversely related to speech rate, longer before tense vowels (particularly tense high vowels), and shorter before voiceless codas. The effect of phonological neighborhood density, while significant, was smaller than any of the phonetic factors examined in the current study. Contrary to previous research, male speakers showed a longer VOT, and the effect of lexical frequency was not significant. The version of vowel tenseness that was most effective was a binary feature which treated low vowels and diphthongs as [+tense].

To our knowledge, the effect of consonant manner in the following syllable (i.e., onset) on VOT has not been previously investigated, as many studies examine monosyllabic words. The findings that VOT is shorter when the next syllable starts with a phonetically voiceless obstruent, and that the effect is greater for plosives than fricatives, reveal that the anticipation of airflow beyond syllable boundaries has a direct influence on articulatory planning on word initial aspiration, and further suggests that the articulatory implementation of phonological planning is likely to involve phonetic details of word-size units (beyond prosodic factors). The fact that flap and glottal stop do not have the same VOT reducing effect is consistent with the idea that the effect is related to planning future airflow requirements.

The reduction in VOT in stops that are followed by voiceless obstruents is similar to Grassmann's Law, in which aspirated consonants in Greek and Sanskrit were de-aspirated when followed by an aspirated consonant in the next syllable (see Collinge, 1985, pp. 47–61). In addition to occurring (apparently separately) in Greek and Sanskrit, similar anticipatory dissimilation of aspiration has been observed in Ofo, an extinct Siouan language (De Reuse, 1981), and four Salish languages (Interior Salish languages Kalispel, Okanagan, Shuswap, and the outlier Salish language Tillamook; Thompson and Thompson, 1985). The fact that anticipatory dissimilation of aspiration has been observed in various unrelated languages suggests a phonetic basis for this pattern. Salmons (1991) argued that Grassmann's Law could have been allophonic at the time that the dissimilation pattern developed in Indo-European languages. This would be consistent with the reduction of aspiration (within the range that is expected for English aspirated stops) that we have observed in /p/ and /k/ preceding voiceless stops and fricatives. A potentially important difference is that the following voiceless obstruents are generally not aspirated, since in most cases in our data they are in unstressed syllables. Nevertheless, if reduction of aspiration occurs in anticipation of voiceless obstruents, even in a language like English where no dissimilation pattern is known, this suggests a phonetic basis for the dissimilation of aspiration.

The connection between liquids and a longer VOT is more mysterious. Since VOT is related to vowel duration, postvocalic liquids may contribute to the perception of vowel duration, although we found that a categorical factor for liquid presence better accounted for the data better than treating the liquid as part of the vowel. Furthermore, syllabic liquids are associated with a longer VOT than would be expected on the basis of their duration. It is also possible that liquids, being produced with a relatively closed vocal tract (at least for a vowel-like sound), have an effect similar to that of tense vowels in delaying the onset of voicing by reducing airflow.

We observed that the following voiceless onset factor is related to speaking rate, i.e., speakers who say the words more quickly within the allotted 2-s interval may have VOTs that are already compressed, resulting in a floor effect such that these factors are not manifested in their speech. If the shorter productions are more representative of spontaneous speech, we may not expect to observe these effects in spontaneous speech.

In addition to phonetic, lexical, and demographic factors, we were able to account for some VOT variation in terms of drift and fluctuation within the course of the task. In general, VOT drifts slightly downward during the reading task, which is consistent with participants getting tired and/or bored. VOT also drifts up and down on a shorter time scale during the task, suggesting changing subglottal pressure, posture, disposition toward the task, or some combination of these and other potential factors. This drift is best accounted for by considering the three previous words' VOT residuals: VOT is longer when recent words had a longer than usual VOT. The residuals account for this drift better than normalized VOT measurements. This means that the VOT (subject to many phonetic factors) is not as relevant as whether the VOT was long or short after taking these factors into account. This is consistent with a variable underlying factor such as changing subglottal pressure, posture, or attitude that interacts with the other factors. Another possibility suggested by a reviewer is that participants are engaging in self-imitation.

The phonetically conditioned VOT effects we observed are anticipatory and therefore require planning (even if the effects have a physiological basis). Figuring out whether the observed VOT drift is cognitive, physiological, or both is more complicated. Looking for a VOT drift effect in natural speech would be one way to explore the underlying cause. If VOT drift is due to self-imitation of recently produced words, we would not expect to see it in natural speech where voiceless stops are less frequent and farther apart (in segmental terms) than in our reading task. If it is due to something observable like seating posture, we would expect to see VOT fluctuations associated with posture in natural speech as well.

The importance of residuals over normalized VOT is also consistent with what has been observed in VOT priming: Levi (2015) found that speakers can be primed to produce a longer VOT in /t/ after hearing a token of *pan* with an unexpectedly long VOT and to produce a shorter VOT in /t/ after hearing a token of *keen* with an unexpectedly short VOT. The short-term (token-by-token) VOT drift that we have observed is superficially similar to medium-term (day-by-day) VOT drift observed by Sonderegger *et al.* (2017, 615), but it is on the time scale of the short-term variation previously observed in VOT accommodation studies (e.g., Nielsen, 2011). It is possible that the short- and medium-term differences are related to the same underlying factor, such as moment-to-moment or day-to-day differences in subglottal pressure, posture, attitude, or something else.

Our results show that realization of VOT is determined by various contextual factors, and that those effects were greater than that of phonological neighborhood density, a lexical factor. Although we observed wide cross-speaker variability in VOT, there was little speaker variability in terms of the realization of those phonetic factors. That is, the ranking of words according to VOT was mostly consistent across speakers, replicating the finding of Allen *et al.* (2003), and the words that our model predicts to have short VOT (e.g., *picture*, *puppy*) were always produced with a VOT near the short end of a given speaker's VOT range. Thus, our data show that the wide variation of within-category VOT is highly structured and therefore predictable within a speaker, based on its phonetic context (as argued in Chodroff and Wilson, 2017 for cross-category VOT). This point has some important implications, particularly when VOT is used as an instrumental measure in psycholinguistic research with a limited number of words (cf. Clark, 1973). For example, the observed structured variation in baseline VOT may help in interpreting differences in VOT accommodation. Nielsen (2011) examined the effect of lexical specificity in spontaneous VOT imitation by comparing the words that participants were exposed to during the exposure phase ("Target words") to the words they were not exposed to ("Novel words"). Her data showed that Target words were produced with a shorter VOT in the baseline recording (as well as in post-exposure recording), potentially confounding the results. Similarly, the highly structured VOT variation needs to be considered in investigating the lexical effects on VOT. For example, Fox *et al.* (2015) showed that word-initial VOT is influenced by the word's phonological neighbors, irrespective of the presence of a minimal pair (MP) differing only in voicing of the initial stop consonant (cf. Baese-Berk and Goldrick, 2009; Kirov and Wilson, 2012; Buz *et al*., 2016). In their experiment, the MP target words (e.g., *peak*) had a MP differing only in voicing of the initial stop consonant (e.g., *beak*), while NMP (non-MP) target words (e.g., *peel*) did not (e.g., *beel*). In their context-neutral condition, the pair *teen-teeth* showed the largest positive MP–NMP difference in VOT (i.e., VOT was greater for *teen* than for *teeth*), while the pair *peak-peel* showed the largest negative MP–NMP difference in VOT. In the former pair, the MP–NMP difference in neighborhood density is also positive, while in the latter pair, it is negative, contributing to the observed effect of neighborhood density. The difference in VOT could be attributed to the lexical competition as the authors argue. However, the patterns of VOT differences in most of the 12 pairs used in the experiment are also predicted based solely on their phonetic contexts, that is, words such as *peel*, *teen*, *cone*, and *comb* are expected to have longer VOTs than their counterparts *peak*, *teeth*, *coast*, and *coat*.

V. CONCLUSIONS

We have shown that VOT is longer in English word-initial voiceless stops that are followed by syllabic or postvocalic liquids (including those that are typically analyzed as onsets), and that VOT is shorter when the next syllable starts with a voiceless obstruent, especially when it is a stop or affricate. Our observations also support earlier findings that VOT is longer in /k/, directly related to following vowel duration, inversely related to speech rate, longer before tense vowels, and shorter before voiceless codas. We also saw that VOT generally drifts downward during the task, but that it also fluctuates in a way that can be partially predicted on the basis of recent productions. These results suggest that within-category VOT variation is highly structured by the phonetic context, similar to the highly structured cross-category variations of VOT reported in Chodroff and Wilson (2017), and should be carefully considered when VOT is used as an instrumental measure in psycholinguistic research.

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[1]*Picture* and *coolant* were the most extreme words in Nielsen's (2011) experiment 1.

[2]The task was a part of a blocked imitation paradigm, following the methods of Nielsen (2011) with minor modifications. Participants read a word list containing many /p/- and /k/-initial words before and after listening to a recording in which the VOT of all word-initial /p/'s had been increased. Only the baseline recording (prior to listening to the recording) was used as the data in the current study.

[3]We are using the mean vowel duration for each word because there is a trading relation between VOT and vowel duration when it is measured on a token-by-token basis, i.e., delaying the onset of voicing in a particular token can simultaneously increase VOT and decrease vowel duration.

[4]This includes all factors from Chodroff and Wilson (2017) that are relevant for our data.

[5]The absence of a lexical frequency effect is a little surprising. To see if the frequency effect was masked or mediated by duration-related factors, we tried a version of model 1 without any of the word duration and vowel duration factors. Lexical frequency is not significant in that model either ($t = 0.166$), or in a model that additionally removes neighborhood density ($t = 0.221$).

[6]Thanks to Robert Port for suggesting to use a factor for the simple presence or absence of a liquid.

[7]For the first word in each speaker's list, we assigned the value of 0 to both the residual factors and the z-scored VOT factors (since no word precedes the first word). The second, third, and fourth words were handled similarly for factors looking back more than one word

[8]The additional random slopes that improve model 8d are as follows: velar (ΔBIC $= -153$), neighborhood density (ΔBIC $= -84$), lexical frequency (ΔBIC $= -49$), vowel duration (by word) (ΔBIC $= -48$), and word list position (ΔBIC $= -35$).

Allen, J. S., Miller, J. L., and DeSteno, D. (**2003**). "Individual talker differences in voice-onset-time," J. Acoust. Soc. Am. **113**, 544–552.

Baayen, R., Piepenbrock, R., and Gulikers, L. (**1995**). "CELEX2 LDC96L14," Linguistic Data Consortium.
Baese-Berk, M., and Goldrick, M. (**2009**). "Mechanisms of interaction in speech production," Lang. Cognitive Process. **24**, 527–554.
Buz, E., Tanenhaus, M. K., and Jaeger, T. F. (**2016**). "Dynamically adapted context-specific hyper-articulation: Feedback from interlocutors affects speakers' subsequent pronunciations," J. Memory Lang. **89**, 68–86.
Cho, T., and Keating, P. A. (**2009**). "Effects of initial position versus prominence in English," J. Phon. **37**, 466–485.
Cho, T., and Ladefoged, P. (**1999**). "Variation and universals in VOT: Evidence from 18 languages," J. Phon. **27**, 207–229.
Chodroff, E., and Wilson, C. (**2017**). "Structure in talker-specific phonetic realization: Covariation of stop consonant VOT in American English," J. Phon. **61**, 30–47.
Clark, H. H. (**1973**). "The language-as-fixed-effect fallacy: A critique of language statistics in psychological research," J. Verbal Learn. Verbal Behav. **12**, 335–359.
Cole, J., Choi, H., Kim, H., and Hasegawa-Johnson, M. (**2003**). "The effect of accent on the acoustic cues to stop voicing in radio news speech," in *Proceedings of the 15th International Congress of Phonetic Sciences*, edited by M.-J. Solé, D. Recasens, and J. Romero, Barcelona, Spain, pp. 15–18.
Collinge, N. E. (**1985**). *The Laws of Indo-European* (John Benjamins, Amsterdam).
De Reuse, W. J. (**1981**). "Grassmann's law in Ofo," Intl. J. Am. Ling. **47**, 243–244.
Fischer-Jørgensen, E. (**1954**). "Acoustic analysis of stop consonants," Le Maître Phonétique **32**, 42–59.
Fox, N. P., Reilly, M., and Blumstein, S. E. (**2015**). "Phonological neighborhood competition affects spoken word production irrespective of sentential context," J. Memory Lang. **83**, 97–117.
Goldinger, S., and Summers, W. V. (**1989**). "Lexical neighborhoods in speech production: A first report," in *Research on Speech Perception Progress Report*, Bloomington, pp. 331–342.
Kessinger, R. H., and Blumstein, S. E. (**1998**). "Effects of speaking rate on voice-onset time and vowel production: Some implications for perception studies," J. Phon. **26**, 117–128.
Kirov, C., and Wilson, C. (**2012**). "The specificity of online variation in speech production," in *Proceedings of the Cognitive Science Society*, Vol. 34(34).
Klatt, D. H. (**1975**). "Voice onset time, frication, and aspiration in word-initial consonant clusters," J. Speech, Lang. Hear. Res. **18**, 686–706.
Koenig, L. L. (**2000**). "Laryngeal factors in voiceless consonant production in men, women, and 5-year-olds," J. Speech, Lang., Hear. Res. **43**, 1211–1228.
Levi, S. V. (**2015**). "Generalization of phonetic detail: Cross-segmental, within-category priming of VOT," Lang. Speech **58**, 549–562.
Lisker, L., and Abramson, A. S. (**1967**). "Some effects of context on voice onset time in English stops," Lang. Speech **10**, 1–28.
McCrea, C. R., and Morris, R. J. (**2005**). "The effects of fundamental frequency level on voice onset time in normal adult male speakers," J. Speech, Lang., Hear. Res. **48**, 1013–1024.
Mielke, J., Nielsen, K., and Magloughlin, L. (**2013**). "Phonetic imitation by individuals with Autism Spectrum Disorders: Investigating the role of procedural and declarative memory," Proc. Mtgs. Acoust. **19**, 060142.
Miller, J. L., Green, K. P., and Reeves, A. (**1986**). "Speaking rate and segments: A look at the relation between speech production and speech perception for the voicing contrast," Phonetica **43**, 106–115.
Nelson, N. R., and Wedel, A. (**2017**). "The phonetic specificity of competition: Contrastive hyperarticulation of voice onset time in conversational English," J. Phon. **64**, 51–70.
Nielsen, K. (**2008**). "Word-level and feature-level effects in phonetic imitation," UCLA dissertation.
Nielsen, K. (**2011**). "Specificity and abstractness of VOT imitation," J. Phon. **39**, 132–142.
Nusbaum, H. C., Pisoni, D. B., and Davis, C. K. (**1984**). "Sizing up the Hoosier mental lexicon: Measuring the familiarity of 20,000 words," Res. Speech Percept. Progress Report **10**, 357–376.
Pierrehumbert, J., and Talkin, D. (**1992**). "Lenition of /h/ and glottal stop," Papers in Laboratory Phonology II: Gesture, Segment, Prosody, pp. 90–117.
Port, R. F., and Rotunno, R. (**1979**). "Relation between voice-onset time and vowel duration," J. Acoust. Soc. Am. **66**, 654–662.
Ryalls, J., Zipprer, A., and Baldauff, P. (**1997**). "A preliminary investigation of the effects of gender and race on voice onset time," J. Speech, Lang., Hear. Res. **40**, 642–645.
Salmons, J. (**1991**). "Motivating Grassmann's law," Historische Sprachforschung/Historical Linguistics **104**, 46–51.
Smith, B. L. (**1978**). "Effects of place of articulation and vowel environment on "voiced" stop consonant production," Glossa **12**, 163–175.
Sommers, M. S. (**2002**). Washington University Speech and Hearing Lab Neighborhood Database.
Sonderegger, M., Bane, M., and Graff, P. (**2017**). "The medium-term dynamics of accents on reality television," Lang. **93**, 598–640.
Swartz, B. L. (**1992**). "Gender difference in voice onset time," Perceptual Motor Skills **75**, 983–992.
Thompson, L. C., and Thompson, M. T. (**1985**). "A Grassmann's law for Salish," Oceanic Linguistics Special Publ. **20**, 134–147.
Whiteside, S. P., and Irving, C. J. (**1998**). "Speakers' sex differences in voice onset time: A study of isolated word production," Perceptual Motor Skills **86**, 651–654.
Yao, Y. (**2009**). "Understanding VOT variation in spontaneous speech," in *UC Berkeley Phonology Lab Annual Report*, pp. 29–43.
Yuan, J., and Liberman, M. (**2008**). "Speaker identification on the SCOTUS corpus," in *Proceedings of Acoustics'08*, pp. 5687–5690.